

Standard errors for the supplier-count distribution $\bar{G}_s(K)$

1 Definition

Throughout, $\bar{G}_s(K)$ is the **increment**, not the CDF:

$$\bar{G}_s(K) = \frac{1}{|L_s|} \sum_{l \in L_s} \Pr[K_{ls} = K] = \frac{1}{|L_s|} \sum_{l \in L_s} \binom{N_s}{K} q_l^K (1 - q_l)^{N_s - K}, \quad (1)$$

the share of cells in sector s hosting *exactly* K suppliers, with $K_{ls} \sim \text{Bin}(N_s, q_{ls})$. At $K = 0$ this coincides with the CDF, so the targeted moment $\bar{G}_s(0) = \overline{(1 - q)^{N_s}}$ and the calibration of $\hat{N}_s$ are unchanged; only the $K \geq 1$ panels are redefined.

Why increments and not the CDF. $\hat{N}_s$ is calibrated as the integer solving $\bar{G}_s(0; n) = \hat{G}_s(0)$, so the $K = 0$ level is fit essentially by construction. Under the CDF convention every panel $\Pr[K_{ls} \leq K]$ *contains* that fitted level, and the untargeted panels re-test what is already targeted. The increments isolate the free content — the *shape* of the count distribution, which is what gate V8 asks about — and, as shown below, the $\hat{G}(0)$ noise largely cancels out of them.

Data side. `G_K.csv` stores the cumulative $G = \Pr[K_{ls} \leq K]$. The empirical increment is the difference of consecutive columns, $\hat{p}_s(K) = \hat{G}_s(K) - \hat{G}_s(K - 1)$.

2 The derivative in n : one line, exact

N_s is an integer, so the relevant derivative is the unit first difference. From $\text{Bin}(n + 1, q) = \text{Bin}(n, q) + \text{Bern}(q)$,

$$\Pr[\text{Bin}(n + 1, q) = K] = (1 - q) \Pr[\text{Bin}(n, q) = K] + q \Pr[\text{Bin}(n, q) = K - 1], \quad (2)$$

hence

$$\boxed{D_s^K \equiv \bar{G}_s(K; n + 1) - \bar{G}_s(K; n) = \frac{1}{|L_s|} \sum_{l \in L_s} q_l \left(\Pr[\text{Bin}(n, q_l) = K - 1] - \Pr[\text{Bin}(n, q_l) = K] \right)} \quad (3)$$

with the first term absent at $K = 0$, where (3) collapses to $D_s^0 = -\overline{q(1 - q)^n}$ — verbatim the quantity already computed beside the analytic derivative in `tools.jl`. So D_s^K needs only $\hat{q}$ and $\hat{N}_s$: no new simulation, no digamma, no continuum relaxation.

3 Variance of the fitted increment

Let $a_s^K = \partial \bar{G}_s(K)/\partial \alpha$ (total, along the profiled manifold) and $b_s^K = (\partial \bar{G}_s(K)/\partial T) f_\gamma$. The calibration at $K = 0$ gives, by the implicit function theorem,

$$dN_s = \frac{d\hat{G}_s(0) - a_s^0 d\hat{\alpha} - b_s^0 d\hat{\gamma}}{D_s^0}. \quad (4)$$

Substituting into $d\bar{G}_s(K) = a_s^K d\hat{\alpha} + b_s^K d\hat{\gamma} + D_s^K dN_s$ and writing $r_s^K = D_s^K/D_s^0$,

$$\boxed{d\bar{G}_s(K) = r_s^K d\hat{G}_s(0) + (a_s^K - r_s^K a_s^0) d\hat{\alpha} + (b_s^K - r_s^K b_s^0) d\hat{\gamma}} \quad (5)$$

Each structural channel enters *residualized against its $K = 0$ direction*: only the part of the α (resp. γ) response that $\hat{N}_s$ cannot absorb survives. Then $\text{Var} = L \mathcal{J} L'$ with L read off (5) and $\mathcal{J}$ the joint covariance already assembled for the $\hat{N}_s$ delta method,

$$\mathcal{J} = \begin{pmatrix} \Sigma_{\hat{G}\hat{G}} & \text{Cov}(\hat{\alpha}, \hat{G})' & \Sigma'_{\gamma\hat{G}} \\ \text{Cov}(\hat{\alpha}, \hat{G}) & \text{Var}_{\text{sw}}(\hat{\alpha}) & \text{Cov}(\hat{\alpha}, \hat{\gamma}) \\ \Sigma_{\gamma\hat{G}} & \text{Cov}(\hat{\alpha}, \hat{\gamma})' & \Sigma_{\gamma\gamma} \end{pmatrix}, \quad \text{Cov}(\hat{\alpha}, \hat{G}) = P \Sigma_{\text{data}}[:, G], \quad (6)$$

PSD by construction, as for V_T . **The only empirical input is $\hat{G}_s(0)$** — the one number the calibration consumed — so the $K \geq 1$ bands need no new bootstrap on the data side.

Two limits worth checking.

- $K = 0$: $r_s^0 = 1$ and both residuals vanish, leaving $\text{Var}[\bar{G}_s(0)] = \text{Var}(\hat{G}_s(0)) = [\Sigma_{\text{data}}]_{G_s G_s}$. The α and target channels do not add, they *cancel*: the fitted moment *is* the target, so the current bar — the α channel alone, $\sqrt{a_s^{0'} \text{Var}_{\text{sw}}(\hat{\alpha}) a_s^0}$ — double counts.
- **Clamped sector**: $\hat{N}_s$ sits at a bound, $dN_s = 0$, and the identity flips to $d\bar{G}_s(K) = a_s^K d\hat{\alpha} + b_s^K d\hat{\gamma}$ for every K , target channel absent. So the band is a per-sector switch keyed on **clamped** — the same distinction the figure already stars.

4 The empirical band

For a bar on the *data* increments, Σ_{data} must be extended: its count block is S rows at $K = 0$ and carries nothing for $K \geq 1$. A bar there today would be invented, which is why SEs are currently drawn on the $K = 0$ panel only. That is a data-pipeline change (bootstrap $\hat{p}_s(K)$ alongside $\hat{G}_s(0)$), outside this repo, and **reconcile_sigma_data** must learn the wider block.

A marginal variance per (s, K) suffices for two bands read side by side. The cross-covariance $\text{Cov}(\hat{p}_s(K), \hat{G}_s(0))$ is needed only for a band on the model–data *difference*; under the increment convention it is second-order, which is a further reason to prefer increments over the CDF, where it is $\approx \text{Var}(\hat{G}_s(0))$ and cannot be ignored.

5 Implementation

1. `model_CP.jl`: `gbar_sector_K`, the exact- K analogue of `gbar_sector`. Keep the unbiased convention — with the $\hat{N}_s$ varieties taken as a sample without replacement from the m simulated draws, the hypergeometric expectation $\mathbb{E}[\binom{k}{j} \binom{m-k}{n-j} / \binom{m}{n}]$ is the Binomial pmf *exactly* for every j , so the estimator is unbiased at every K , not only at $K = 0$.

2. $\bar{G}_s(K)$ as a diagnostic block in `compute_moments` (*not* a moment), so the Jacobian has rows for a_s^K and b_s^K .
3. D_s^K from (3), beside the existing `D_firstdiff`.
4. `var_G_curve_delta.npy` from the existing `Joint`; the clamped-keyed switch in `plot_count_curve`.

The first difference is used throughout. The $\hat{N}_s$ delta method previously divided by the *analytic* slope $(1 - \hat{q})^n \ln(1 - \hat{q})$. It now divides by D_s^0 , the same unit first difference as everything else in this file: N_s is an integer, one variety *is* the step, and D_s^0 enters both r_s^K and $\text{Var}(\hat{N}_s)$, so a mixed convention would make the two inconsistent. It is evaluated *through* `gbar_sector` — the difference of the same unbiased estimator the bisection matched on, not of a plug-in that drifts from it by $O(\hat{q})$. The plug-in slope is still printed beside it as a scale check; the published se_N moves slightly as a result.

6 The empirical variance file

`<baseline>/G_K_var.csv`, optional, columns `A129`, `K`, and either `var` or `se` — names matched case-insensitively, as `G_K.csv` is. It must hold the bootstrap variance of the **increment** $\hat{p}_s(K)$, not of the cumulative $\hat{G}_s(K)$: those coincide only at $K = 0$.

The value column is found by name (`var/variance`, else `se/sd/std`); a file written off `G_K.csv`'s own schema often keeps it called `G`, which names the object measured rather than the statistic, so whatever single column remains beside `A129` and `K` is read as a variance and the choice is announced. Two spare columns are refused rather than guessed.

Only $K \leq \text{COUNT_CURVE_K_MAX}$ is kept, from this file and from `G_K.csv` alike, so the variance file need only cover the low K actually plotted.

`A129` holds the sector **codes**, exactly the values `G_K.csv` carries (`C30C`, `2501`, ...) — not a $1 \dots S$ index. Sectors are matched by value, so row order is irrelevant, but the two files must agree on the code's dtype; the loader compares the stripped string form and names any code it could not find rather than returning an all-NaN band silently.

That coincidence is a free consistency gate and is checked rather than assumed. Since $\hat{p}_s(0) = \hat{G}_s(0)$ is block 6 of the moment vector, its bootstrap variance is already the count block of Σ_{data} ; the loader compares the two and says so when they differ by more than 5%, the most likely cause being a file that tabulates the CDF's variance.

Bootstrap the whole curve in one pass, resampling as the existing $\beta \rightarrow \gamma \rightarrow G$ block does. Only the marginal variance per (s, K) is consumed here — the cross-covariance $\text{Cov}(\hat{p}_s(K), \hat{G}_s(0))$ would be needed only for a band on the model–data *difference*, and under the increment convention it is second order.